St Edmund's College
Senior Assessment Information and Conditions

Focus of the task	
Terms/Prior knowledge	Stanford Design Thinking Process, Data Storytelling, social and ethical issues associated with data, variability in data, Data Science Processes
Skills:	Manipulating and analysing data in Excel, Google Colab

The task

Your assignment involves using data to tell a succinct and well-reasoned story for a specific purpose and audience. Inspired by the Dear Data project (Lupi & Posavec), you will collect your own personal data and combine it with data from other sources to prototype and build an interactive data story using Google Colab, CODAP and HTML. You will be expected to document the design thinking you use for this task and assess the effectiveness of your data story.

A good data story does more than present numbers. It uses data to inform, persuade or inspire action. It presents information in a way that is clear and accessible, and it helps the audience connect with what the data actually means. As you work through your chosen topic, keep asking yourself what you want your audience to think, feel or do differently as a result of your story.

You are to choose one of the following topics for your data story:

- Emotions- Pick one emotion and track how often you notice it around you, in conversations, on social media, in the news, over at least 7 days. Use your observations alongside broader datasets to tell a data story that helps your audience understand this emotion in a new way.
- Driver behaviour — Spend at least 4 sessions observing drivers at a location near you. Record what you see. Use your observations alongside road safety and crash data to tell a story that persuades your audience to take the risks of everyday driving more seriously.
- Social Media - Track your own screen time and platform use over at least 4 days. Use your data alongside research on young people and social media to tell a story that helps your audience make sense of Australia's new laws restricting social media access for under 16s.
- Own choice – discuss with teacher

NB You will be expected to use data from at least three separate sources. One of these must be data you have collected yourself. You will also be expected to use 2 different platforms to help you clean, model and analyse your data.

You will be expected to show evidence of your thought processes when completing this task. You must complete two progress check ins prior to submitting your final assignment.

NB AI can only be used in the assessment for brainstorming, creating structures, and generating ideas for improving work. AI cannot be used to generate code.

Useful Resources and Sources for Data

Emotions	Driver Behaviour	Social Media	Own Choice
https://www.kaggle.com/datasets/debarshichanda/goemotions kaggle.com/datasets https://ourworldindata.org/ https://www.abs.gov.au/	https://catalogue.data.infrastructure.gov.au/dataset/australian-road-deaths-database kaggle.com/datasets https://ourworldindata.org/	https://www.esafety.gov.au/key-topics https://www.pewresearch.org/topic/internet-technology/	https://data.gov.au/ https://researchdata.edu.au/ https://www.data.qld.gov.au/dataset https://www.abs.gov.au/

	https://data.gov.au/ https://www.data.act.gov.au/	kaggle.com/datasets https://ourworldindata.org/	https://www.mastersindatascience.org/resources/data-visualization/ https://github.com/awesomedata/awesome-public-datasets https://www.data.act.gov.au/ https://www.kaggle.com/datasets https://www.sports-reference.com/ https://www.espncricinfo.com/records

Task Instructions
Step 1: **Empathise and Ask Questions** – think about your topic and the people impacted by it. What questions come to mind? What are their needs? Create a user persona and write a minimum of 3 questions your data story will attempt to answer.
Step 2: **Define** – Define the problem statement from the user's perspective.
Step 3: **Ideate and Gather and Organise Data** - brainstorm approaches to collecting and presenting your data. Start collecting. Your self-collected data must span a minimum of 4 time periods.
Step 4: **Prototype and Model** - start creating charts to model your data using at least two different platforms. Include a minimum of 4 charts, one of which must include multivariate data. Prototype your story.
Step 5: **Test and Analyse** - synthesise your findings and refine your modelling. Document in dot points how you have examined trends and variability, any social and ethical issues the data raises, and whether the data includes inherent bias.
Step 6: **Create your Story -** build your data story as a single HTML page with four charts. Include a hand-drawn Dear Data postcard of your self-collected data with a legend, photographed and embedded in the page. Get feedback and refine. Use in-text citations. Remember, your story is not a description of your charts. Use the data to argue, persuade or inspire. Include a Reference List at the end of your story.
Step 7: Present your Story and Assess its Effectiveness – Create and deliver a presentation that effectively showcases your story and evaluates how successful it is in: • Answering the questions posed in Step 1 • Addressing the needs of your user persona • Addressing the problem statement from Step 2 • Presenting data to support the story, including your choice of chart types, data cleaning and prototyping • Using data to argue, persuade or inspire rather than simply describe, and addressing trends, variability, bias and ethical issues You will be expected to answer questions about your presentation.
Step 8: Make sure you have completed your progress checks
Step 9: Submit your files. This includes a transcript of your presentation.

Assessment criteria
You will be assessed on the degree to which you demonstrate: • Knowledge and Understanding • Skills Please see the Rubric for a complete breakdown of how marks will be awarded.

Submission requirements
You will be expected to submit the following: • Data Story files and PowerPoint with script for presenation • Files used in programs such as CODAP to create chart

Criteria	**Outstanding Achievement**	**High Achievement**	**Satisfactory Achievement**	**Partial Achievement**	**Limited Achievement**
Data Story – Content 20 marks	Story is expertly crafted and uses data to argue, persuade or inspire in a way that is compelling and clearly targeted at its purpose and audience. The Dear Data postcard is creative, meaningful and integrates seamlessly with the story.	Story is effectively crafted and uses data to argue, persuade or inspire. Purpose and audience are clear throughout. The Dear Data postcard is well considered and connects to the story.	Story is satisfactorily crafted and makes reasonable attempts to use data to argue, persuade or inspire. Purpose and audience are mostly clear. The Dear Data postcard is included and connects to the data.	Story describes the data more than it uses it to argue, persuade or inspire. Purpose and audience are considered but inconsistently applied. The Dear Data postcard is included but its connection to the story needs more development.	Story largely describes the data with little attempt to argue, persuade or inspire. Purpose and audience are unclear. The Dear Data postcard is missing or has little connection to the story.
Data Story – Skills 20 marks	Demonstrates excellent skills in gathering, cleaning, presenting, analysing and synthesising data. Self-collected data is thorough and well documented. Charts are very well constructed, appropriate for purpose and extend the story. Excellent use of two or more platforms.	Demonstrates good skills in gathering, cleaning, presenting, analysing and synthesising data. Self-collected data is well documented. Charts are well constructed and appropriate for purpose. Good use of two or more platforms.	Demonstrates satisfactory skills in gathering, cleaning, presenting, analysing and synthesising data. Self-collected data is included and documented. Charts are satisfactorily constructed and appropriate for purpose. Two platforms used.	Demonstrates developing skills in gathering, cleaning, presenting and analysing data. Self-collected data is included but documentation is limited. Charts show developing skills. Evidence of using two platforms but application is inconsistent.	Demonstrates limited skills in gathering, cleaning, presenting and analysing data. Self-collected data is minimal or poorly documented. Charts show limited skill. Limited evidence of using two platforms.
Data Story – Construction 10 marks	HTML page is very well constructed. Visuals and text work together cohesively to create a polished and engaging page. Excellent use made of CSS. Postcard is well presented and the legend is clear. In-text citations are used correctly throughout.	HTML page is well constructed. Visuals and text work well together. Good use made of CSS. Postcard and legend are clear. Most in-text citations are used correctly.	HTML page is satisfactorily constructed. Visuals and text are combined adequately. Satisfactory use made of CSS. Postcard and legend are included. Some in-text citations are used.	HTML page is functional but visuals and text are not well integrated. CSS used but with varied effect. Postcard or legend is unclear. Limited use of in-text citations.	HTML page shows limited construction skills. Visuals and text are poorly integrated. Limited use made of CSS. Postcard or legend is missing. Little or no attempt at in-text citations.

Analysis and Presentation– 25 marks	Presentation is confident and engaging and effectively showcases the story. Evaluation of the story is insightful and addresses all five criteria with depth and clarity. Demonstrates an excellent understanding of trends, variability, bias and ethical issues and how they shaped the story. Answers questions with confidence and demonstrates a thorough understanding of all aspects of the work.	Presentation is confident and effectively showcases the story. Evaluation addresses all five criteria well. Demonstrates a good understanding of trends, variability, bias and ethical issues. Answers questions well and demonstrates a good understanding of the work.	Presentation is satisfactory and covers the story adequately. Evaluation addresses most of the five criteria. Demonstrates a sound understanding of trends, variability and bias. Answers most questions adequately and demonstrates a sound understanding of the work.	I Presentation covers the story but lacks depth in places. Evaluation addresses some of the five criteria but others need more development. Some understanding of trends, variability and bias is evident. Answers some questions but struggles with others.	Presentation is limited in its coverage of the story. Evaluation addresses few of the five criteria. Limited understanding of trends, variability and bias is evident. Struggles to answer questions about the work.
Design Thinking	Questions show considerable thought and imagination and clearly drove the story. User persona is very well constructed and demonstrates an excellent understanding of the needs of the audience. Problem statement is succinct and very well defined. Prototypes show excellent progression of ideas and clear evidence of refinement. Microsoft Form is very well constructed and feedback is effectively used to enhance the story.	Questions show good thought and imagination. User persona is well constructed and demonstrates a good understanding of the needs of the audience. Problem statement is well defined. Prototypes show good progression of ideas and evidence of refinement. Microsoft Form is well constructed and feedback is used to enhance the story.	Questions show satisfactory thought. User persona is satisfactorily constructed and demonstrates a sound understanding of the needs of the audience. Problem statement is satisfactorily defined. Prototypes show satisfactory progression of ideas. Microsoft Form is satisfactorily constructed and feedback is used to enhance elements of the story.	Questions show some thought but lack depth. User persona is included but needs more detail to demonstrate a sound understanding of the audience. Problem statement lacks clarity. Prototypes show some progression but refinement is limited. Microsoft Form is satisfactorily constructed but feedback does not appear to have been used to improve the story.	Questions show limited thought. User persona needs considerably more detail. Problem statement is poorly defined. Prototypes show little progression of ideas. Microsoft Form is either missing or poorly constructed. Limited evidence that feedback has been used to improve the story.
Reference List, In-text citations and Progress checks	Excellent variety of reliable sources. Citations and Reference List are correctly	Good variety of reliable sources. Citations and Reference List are mostly	Satisfactory variety of sources. Satisfactory attempt at citations and	More than three sources but variety is limited. Developing attempt at	Limited variety of sources. Limited attempt at citations and

5 marks	formatted in APA style. All three progress checks submitted and demonstrate excellent commitment to the task.	correctly formatted in APA style. All three progress checks submitted and demonstrate good commitment.	APA style Reference List. At least two progress checks submitted	citations and Reference List. At least one progress check submitted.	Reference List. Progress checks largely missing.